4.7

PAEDIATRICS

Gastrointestinal Infections

CHAPTER 4.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Acute Gastroenteritis			
Usually viruses: *Rotavirus, Norovirus*	Antibiotics are not recommended		
Dysentery			
Shigella *E. coli* *Campylobacter*	**Mild, uncomplicated:** Antibiotics not recommended		• Campylobacter: Macrolides (azithromycin, clarithromycin) is preferred. • Caution of increasing resistance to ciprofloxacin.
Severely ill, malnourished, immunocompromised	Ceftriaxone 50 mg/kg/dose IV q12-24H (max 2 g daily) for 5 days	**< 3 months old:** Ampicillin 25 mg/kg/dose PO/IV q6H for 5-7 days	

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	Recommended	**Alternative**	
		Ciprofloxacin 15 mg/kg/ dose IV q12H for 3 days **OR** Azithromycin 10 mg/kg/ dose IV q24H for 3 days	
Typhoid fever			
Salmonella typhi *S. paratyphi A & B*	**Mild, uncomplicated:** Azithromycin 15-20 mg/kg/dose (max 500 mg) PO q24H for 7 days	**Mild, uncomplicated:** Ciprofloxacin 20 mg/kg/ dose (max 750 mg) PO q12H for 7 days	• Bacteraemia and severe or complicated enteric fever should be treated for 10-14 days. • Delay in resolution of fever is common; may not occur until 3-5 days after starting antibiotic treatment even if the organism is fully susceptible. • In a stable child, high dose oral antibiotics may be started to complete the course once the fever has resolved. • *Check G6PD status prior to starting Co-trimoxazole.
	Severe, complicated: Ceftriaxone 50 – 75 mg/kg/dose IV q24H	**Severe, complicated:** Ampicillin 50 mg/kg/ dose IV q6H **OR** Ciprofloxacin 15 mg/kg/dose IV q12H for 5 days	

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	Chronic carrier: Amoxicillin 30 mg/kg/dose (max 1g) PO q8H for 28 days	**Chronic carrier:** Co-trimoxazole 4-5 mg/kg/dose of Trimethoprim (max 320 mg) PO q12H* **OR** Ciprofloxacin 20 mg/kg/dose (max 750 mg) PO q12H	
Peritonitis			
Primary peritonitis *S pneumonia* *Streptococcus pyogenes* *Staphylococcus aureus* *Escherichia coli* *Klebsiella pneumoniae*	Ceftriaxone 50 mg/kg/dose (max 2 g) IV q12H **OR** Cefotaxime 50 mg/kg/dose (max 2 g) IV q6-8H + Metronidazole 15 mg/kg/dose IV loading dose then 7.5 mg/kg/dose IV q8H (max 4 g daily)	Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) IV q8H **OR** Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H (max 16 g daily)	

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Secondary peritonitis *E coli* *Klebsiella* *Enterococcus* *Enterobacter* *Pseudomonas* *Proteus spp* *Anaerobes Bacteroides* *Candida spp* (rare)	Ceftriaxone 50 mg/kg/ dose (max 2 g) IV q12H **OR** Cefotaxime 50 mg/kg/ dose (max 2 g) IV q6-8H + Metronidazole 7.5 – 10 mg/kg IV q8H	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H **OR** Meropenem 20 mg/kg/dose IV q8H ± Vancomycin 15 mg/kg/dose q8H	• Peritonitis secondary to gut perforation is often polymicrobial, which may include *Candida spp.* • Consider adding an antifungal in critically ill patients. • Documented non-VRE and *Enterococcus faecalis*: Vancomycin 15 mg/kg/dose IV q8H. • Piperacillin/Tazobactam has *Enterococcus spp* coverage.
Liver Abscess			
Pyogenic hepatic abscess *E coli* *Enterococcus spp.* *Klebsiella* *Pseudomonas* *Proteus* *Bacteroides* *Streptococcus milleri* *Staphylococcus aureus*	Ceftriaxone 50 mg/kg/ dose (max 2 g) IV q12H + Metronidazole 7.5 - 10 mg/kg/dose IV q8H (max 4 g daily)	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H	• Often associated with multiple abscesses within the spleen. • Duration: 4-6 weeks.

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Cholangitis (Biliary Atresia)			
Escherichia coli *Klebsiella* *Enterococcus* *Enterobacter* *Pseudomonas* *Proteus* *Bacteroides*	Cefotaxime 50 mg/kg/ dose (max 2 g) IV q6-8H **OR** Ceftriaxone 50 mg/kg/ dose (max 2 g) IV q12H + Metronidazole 7.5 – 10 mg/kg IV q8H	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H	• Duration: 10-14 days.
Cholecystitis			
Escherichia coli *Klebsiella* *Enterococcus* *Enterobacter* *Pseudomonas* *Proteus* *Bacteroides*	Ceftriaxone 50 mg/kg/ dose IV q12H + Metronidazole 7.5 mg/kg/dose IV q8H	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H	• Majority are acalculous; associated with other systemic diseases like typhoid, *Streptococcal* infection, *S aureus.*

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Catheter-related Bloodstream Infection (Home TPN)			
S aureus, CONS, *Escherichia coli,* Klebsiella, Pseudomonas	Ceftazidime 50 mg/kg/dose IV q8H + Vancomycin 15 mg/kg/dose IV q8H		